higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL CONSTRUCTION N4

(8060004)

27 November 2019 (X-Paper)
09:00–12:00

REQUIREMENTS: ONE A2 drawing sheet

This question paper consists of 5 pages and 1 diagram sheet.

Copyright reserved

DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING AND STRUCTURAL CONSTRUCTION N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Strictly adhere to the following:

   Do QUESTIONS 1, 2 and 3 on the front of the DRAWING SHEET.
   Do QUESTIONS 4, 5 and 6 on the back of the DRAWING SHEET.
   Neatly print QUESTION 7 on the DRAWING SHEET.

   Do NOT use an ANSWER BOOK.

2. Read ALL the questions carefully.

3. Number the answers according to the numbering system used in this question paper.

4. Do ALL drawings, labels and written answers in pencil with bold outlines.

5. The drawings must be done in accordance with National Standards and be fully labelled with descriptive notes, dimensions and titles.

6. Maintain a balanced layout.

7. Work neatly.

Copyright reserved

QUESTION 1

A one-and-a-half brick foundation wall is supported on a 700 × 330 mm thick concrete foundation of which the top is 1 125 mm below the natural ground level. A 100 mm thick concrete basement floor is cast on top of a 75 mm thick oversite concrete bed resting on top of the concrete foundation on the inside of the basement. A sheet of damp-proof membrane is sandwiched between the two concrete layers.

The ground floor is constructed of 150 mm × 50 mm floor joists with 100 mm × 22 mm tongue and grooved floorboards supported by a 114 × 75 mm wall plate resting on the inside wall THREE brick courses above the natural ground level. The floor is completed with a 100 × 22 mm timber skirting and 19 mm quadrant.

A 150 × 220 mm air vent, 300 mm above ground level, is built into the wall to ventilate the timber floor and basement below. The basement wall is reduced to a 270 mm cavity wall THREE courses above the ground level.

Draw, to scale 1:10, a vertical section through the basement wall to show the following:

- 330 mm wide basement reduced to a 270 mm cavity wall
- Concrete and timber floors
- Natural ground level
- Air vent
- Plaster, skirting and quadrant on the inside of the cavity wall only
- One tie wire
- Damp-proof course 150 mm from the natural ground level as well as the vertical damp-proofing

Include ALL dimensions, labels and hatching key symbols on the completed drawing. **[24]**

QUESTION 2

A two-ringed rough-brick segmental arch must be built over a fireplace. The span of the arch must be 1 200 mm with the rise one sixth of the span.

Draw, to scale 1:10, the front view of the segmental arch. Draw the bricks on the right-hand side and show the geometrical layout on the left-hand side. The completed drawing must include labels and dimensions. Clearly indicate ALL construction lines. **[12]**

Copyright reserved

QUESTION 3

Draw, to scale 1:10, the alternate plan views of a one-and-a-half brick corner built in Flemish bond. The internal dimensions for the vertical part is 715 mm and 880 mm for the horizontal section. Show ONE stop end each on the horizontal and vertical parts. Include ALL dimensions and labels. Provide a title and scale for the drawing.

NOTE: The TWO views must be drawn one above the other. **[10]**

QUESTION 4

The DIAGRAM SHEET (attached) shows an isometric view of a 254 × 146 × 31,3 kg/m secondary steel beam intersecting with a 406 × 178 × 53,8 kg/m primary beam with both beams level and in line at the top. The TWO beams will be connected by means of 70 × 70 × 6 mm rolled steel angles and diameter-16 snap-head rivets.

Draw, to scale 1:5, the following views:

4.1 A cross-sectional front view through the secondary steel beam

4.2 A cross-sectional front view through the primary steel beam

The completed drawings must include ALL labels and hatchings. Provide a title and scale to each of the drawings. **[21]**

QUESTION 5

Draw, to scale 1:2, a vertical section through a domestic side-hung steel window frame to show the top section, the glazing bar and the bottom section of the window. **[10]**

QUESTION 6

The ceiling of a dwelling must be dropped 450 mm below the tie beam of a roof truss to form a suspended ceiling. The ceiling will be constructed of 75 × 38 mm ceiling joists nailed to 38 × 38 mm batten hangers at 400 mm centres. The batten hangers are also nailed to the tie beam at 400 mm centres. The ceiling is completed using 6 mm Rhino ceiling board nailed to 75 × 38 mm ceiling joists. A 75 mm cove cornice will close the gap at the wall and ceiling.

Draw, to scale 1:10, a cross-sectional view of the suspended ceiling to show the one-brick wall, part of the roof truss, the construction details of the ceiling and 19 mm plaster to the inside of the wall. **[15]**

Copyright reserved

QUESTION 7

Window frames are set up by following a set sequence of steps. The list below shows EIGHT steps that must be followed before the bricklayer can build the frame into the wall.

Rewrite these steps in the correct sequence:

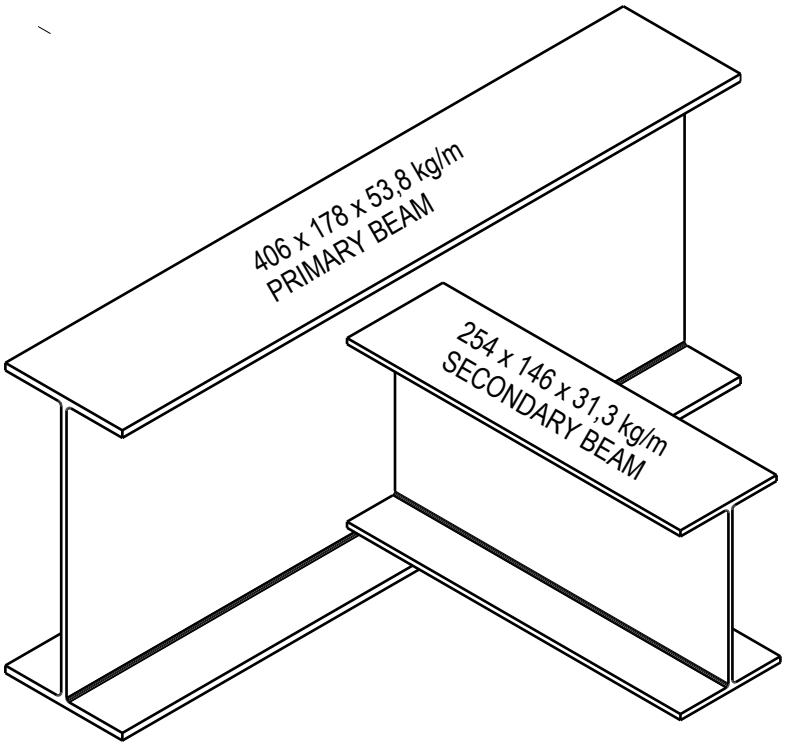
- Place two bricks, on edge, on top of the wall 100 mm inside the marked positions of the window.
- Place the frame on the marked positions on the wall.
- Proceed with levelling the frame using folding wedges on top of the bricks-on-edge and plumbing the frame by moving the stays forward or backwards.
- Mark out the position of the window frame on the wall.
- Place a sole plate on the inside concrete floor and weigh down with bricks or concrete blocks to keep the stays in position.
- Place the window on top of the bricks.
- The window is now ready to be built in.
- Drive a nail halfway into the top of two timber stays then hook the stay onto the top of the window frame to hold the frame in position. **[8]**

TOTAL: 100

Copyright reserved

(8060004)

DIAGRAM SHEET



Copyright reserved